

# Muhammad Tegar Abhiram

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<https://github.com/garr007>

## DESCRIPTION

Information Technology graduate from Universitas Brawijaya specializing in AI Engineering and Data Science. Proven track record designing end-to-end computer vision pipelines (YOLOv8/v11) and building workflows using n8n orchestration. Adept at optimizing LLM prompt architectures, reducing latency, and engineering scalable data pipelines using Python. Experienced in translating complex datasets into production-grade predictive models and interactive analytics dashboards.

## EDUCATION

### Brawijaya University

Faculty of Computer Science; Majoring Information Technology  
Cumulative GPA: 3.8/4.0

Aug 2021 – Aug 2025

Malang, Indonesia

## WORK EXPERIENCE

### PT. Mitra Graha Integrasi (MIGRASI)

Nov 2025 – Mei 2026

Data Analyst & AI Engineer Intern

Bogor, Indonesia

- Engineered a semantic Vector Search system utilizing BERT embeddings and a FAISS vector database, achieving 80% search accuracy
- Refactored prompt architectures to accelerate system response times, cutting inference latency by 50% across production workflows
- Optimized LLM prompt architecture and token utilization, reducing API latency by 50% and decreasing prompt character overhead by 79%
- Designed and deployed an autonomous AI chatbot workflow using n8n, integrating LLM APIs into the company's production platform to automate real-time customer support inquiries

### Badan Riset dan Inovasi Nasional (BRIN)

Feb 2024 – Jul 2024

Computer Vision Research Intern

Bandung, Indonesia

- Engineered a custom YOLOv8 object detection model for real-time Mouse Movement Detection, achieving 99% accuracy; documented methodologies and co-authored a scientific paper on the research findings
- Developed and optimized a YOLOv8 instance segmentation model for Tuna Fish Classification to automate quality assessment, achieving 80% accuracy
- Curated, annotated, and pre-processed a computer vision dataset of 1,500+ images utilizing Roboflow, implementing custom augmentation pipelines to drastically reduce model overfitting

## ORGANIZATIONAL EXPERIENCE

### Keluarga Besar Mahasiswa Departemen Sistem Informasi FILKOM UB

Jan 2023 – Jan 2024

General Secretary of Badan Perwakilan Mahasiswa Departemen Sistem Informasi

Malang, Indonesia

- Spearheaded cross-functional alignment within a 7-member core committee to execute high-impact organizational initiatives smoothly
- Drafted and instituted 5+ foundational regulations to govern organizational compliance, leading oversight for all major executive activities
- Directed the planning, budgeting, and performance evaluation of 10+ major community events, successfully driving engagement and operational success

### Faculty of Computer Science (FILKOM) University of Brawijaya

Aug 2023 – Dec 2023

Algorithm and Data Structure Laboratory Asisstant

Malang, Indonesia

- Mentored 42 students in core computer science concepts, delivering technical code walk-throughs on advanced data structures, sorting algorithms, and time-complexity optimization
- Evaluated and code-reviewed student programming assignments, providing constructive debugging feedback and grading implementations based on efficiency and algorithmic correctness

## House of Technology FILKOM UB

Head of IT Paper & Poster Division

May 2022– Nov 2022

Malang, Indonesia

- Led a 5-member executive team in planning and executing the annual IT Paper and Poster Exhibition
- Attracting 10+ participants and showcasing innovative tech projects from students across the university
- Managed the final stage event, coordinating activities for 5 finalists to ensure a smooth and successful event

## PROJECTS

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### AI Automation Project: Mira Assistant – Enterprise RAG Workflow Optimization

Mar 2026 – May 2026

Developed an end-to-end autonomous customer support agent and dynamic knowledge-base pipeline using n8n orchestration and JavaScript logic

Bogor, Indonesia

- Engineered an automated data ingestion pipeline using n8n and custom JavaScript, scheduling periodic document downloads from cloud storage via automated triggers
- Integrated a secure CMS API Sync Pipeline with admin authentication, enabling automated synchronization of new enterprise knowledge into the core RAG ingestion system
- Designed a custom text-processing pipeline that chunks documents at chunk\_size=1000 and generates embeddings using OpenAI's text-embedding-3-small for storage in a Pinecone Vector Store
- Built an AI Agent using GPT-4o-mini with a Window Buffer Memory and strict conditional routing to ensure accurate, secure, and hallucination-free contextual responses

### Computer Vision Project: Mouse Movement Detection

Feb 2024 – Jun 2024

Developed a Computer Vision application for detecting and tracking mouse movement using YOLOv11

Malang, Indonesia

- Curated and pre-processed a specialized dataset of 300+ images via Roboflow, training a custom YOLOv11 object detection model to achieve a 99% tracking accuracy rate
- Engineered an interactive Graphical User Interface (GUI) to visualize real-time bounding boxes and inference tracking results seamlessly for end-users.

### Analyzing Customer Churn Using Tableau: Databel Telecom Company by DataCamp

Nov 2023 – Dec 2023

Data Analyst and Visualization using Tableau

Malang, Indonesia

- Conducted exploratory data analysis on telecom customer data, engineering custom calculated fields and defining strategic KPIs to quantify driver metrics behind customer churn
- Developed an interactive suite of 4 Tableau dashboards and a data story, delivering actionable insights to identify high-risk customer segments and optimize retention strategies

## SKILLS AND CERTIFICATIONS

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### HARD SKILLS

- Programming Languages: Python (Advanced), R (Intermediate), SQL
- AI & Computer Vision: Ultralytics YOLO (v8/v11), OpenCV, TensorFlow, Neural Networks, SVM
- Data Engineering: ETL (Pentaho, Talend), Database Management Systems (DBMS), MySQL
- Analytics & BI: Data Analysis & Visualization, Tableau, Streamlit, Exploratory Data Analysis (EDA)
- Workflows & MLOps: n8n Orchestration, Image Curation & Pre-processing (Roboflow), Graphical User Interfaces (GUIs)

### Certifications & Training:

- DBS Foundation Coding Camp 2024 (Dicoding) – Machine Learning Developer. ([see certificate](#))
- Indosat Ooredoo Hutchison Digital Camp 2024 (Dicoding) - Data Scientist. ([see certificate](#))
- Analyzing Data in Tableau; Creating Dashboards in Tableau; Case Study: Analyzing Customer Churn in Tableau by DataCamp. ([see certificate](#))
- Statistics using R for Data Science; Data Preparation in Data Science Using R; Data Visualization in Data Science using R; Advanced Data Visualization with ggplot2 using R by DQLab. ([see certificate](#))
- SQL (Advanced) Certificate; Python (Basic) Certificate by HackerRank. ([see certificate](#))
- EF SET English Certificate 83/100 (C2 Proficient). ([see certificate](#)).

### Languages:

- Indonesia (Native)
- English (Fluent)